

C. V. RAMAN GLOBAL UNIVERSITY, BHUBANESWAR
MID TERM EXAMINATION-2026
2nd SEMESTER B.Tech. ALL

SUBJECT: Introduction to Electronics Engineering (IEE)

SUBJECT CODE: EC101

TIME: 1 hours

FULL MARK: 20

BRANCH: 2nd Sem, B.Tech.

1. Answer all questions each carrying equal marks [2 × 5 = 10]

- f) Differentiate between a conductor and a semiconductor based on their energy band structure.
- g) State the difference between Zener and Avalanche breakdown.
- h) What is the depletion region in a PN junction and how is it formed?
- i) Explain the Early effect in transistor.
- j) For a given value of $\alpha = 0.98$, find β .

3. Answer the questions. [5 × 2 = 10]

- a. Using a suitable PN junction diode circuit, explain the operation of a PN junction diode under forward bias and reverse bias conditions. Illustrate the behavior with the help of the V-I characteristics curve.

OR

Explain the working principle of a full wave rectifier with a neat circuit diagram. Draw and explain the input and output waveforms, and state the value of the ripple factor.

- b. Draw a neat circuit diagram of a transistor in the common Emitter mode.
i) Explain working principle, ii) sketch its output characteristics and indicate the 3 different regions, and iii) sketch its input characteristics.

OR

Explain the construction and working principle of a JFET.

COS	QUESTIONS	TOTAL MARKS
CO1:	1(a, b, c), 2(a)	2, 2, 2, 5
CO2:	1(d, e) 2(b)	2, 2, 5